Dr Caitlin D. Kuempel

Senior lecturer · Griffith University, Brisbane, Australia
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Professional Highlights

1. Proven conservation program leader with over a decade of experience delivering applied research and management partnerships across the Pacific and Australia, focused on marine spatial planning, coral reef conservation, watershed management, and sustainable development.
2. Track record in project design and delivery: secured and managed over $1.2 million in project funding in the past two years, leading multi-institutional collaborations with governments, NGOs, and industry.
3. Strong implementation and coordination experience: served as Monitoring and Data Management Lead for WWF's Coral Reef Rescue Initiative and currently lead the Cumulative Impacts Workstream for the Blue Economy Cooperative Research Centre Marine Spatial Planning project.
4. Skilled communicator and collaborator: experienced in engaging diverse stakeholders—from local communities to global NGOs—to co-develop data-driven conservation solutions.
5. Capacity building and leadership: trained over 300 professionals and students in R programming and decision-support tools such as Marxan; mentor to early-career scientists and conservation practitioners.
6. Recognised communicator and advocate: research has been featured in The Guardian, The Washington Post, Time magazine, and The Conversation (>200,000 reads); finalist in national science communication competitions and Queensland Women in STEM Award 2025.
7. Committed to inclusive leadership and organisational excellence: lead of the Sustainable Seas Research group, active participant in Griffith's Women in Leadership Program and representative on the School's Early Career Researcher Committee.

Education

PhD Conservation Science, The University of Queensland, St. Lucia, QLD, Australia — Sept. 2018

Thesis: Evaluating trade-offs between quantitative and qualitative protected area objectives Supervisors: Prof. Hugh Possingham, Prof. Catherine Lovelock and Dr. Alienor Chauvenet Dean's Award for Outstanding HDR thesis (ranked within top 10% in field by examiners).

M.S. Marine Biology Northeastern University, Nahant, MA — Dec. 2013

Coral Reef Ecology (GPA 4.0/4.0), Supervisor: Dr Andrew ALtieri

BSc Environmental Science (Honours), BA French, University of Iowa, Iowa City, IA — May 2012

Honors in coral morphometrics, Certificate in Sustainability

Professional Experience

Senior Lecturer, Marine Ecosystem Modelling, Griffith University, Brisbane, Australia — June 2022-present

Balanced profile: 40% teaching, 40% research, 20% service

Post-doctoral Fellow, ARC CoE for Coral Reef Studies, UQ, Brisbane, Australia Feb. 2020-May 2022

Explored ways to manage and monitor climate resilient reefs to benefit people and nature as part of the World Wildlife Fund's Coral Reef Rescue Initiative

Post-doctoral Fellow, NCEAS, Santa Barbara, CA Sept. 2018-Dec. 2019

Quantified, compared and mapped the production and environmental impacts of all major food systems as part of a collaborative working group

Conservation Planning Consultant, IUCN and The University of Queensland July-Aug.2018

Collated data and developed conservation plans for four Pacific Island nations using the spatial conservation planning software, Marxan.

Research Assistant: Smithsonian Tropical Research Institute, Bocas del Toro, Panama April-Oct. 2014

Thalassia testdinum Experimental Network (TEN): investigated the impacts of nutrients and grazer abundance on seagrass across a latitudinal gradient (Florida, Belize, Panama).

Conservation Intern: Central Caribbean Marine Institute, Little Cayman, Cayman Islands Jan.-April 2014

Assisted with lionfish management research, coral nursery, and climate data collection. Provided weekly educational talks and tours to guests and tourists.

Short-Term Fellow: Smithsonian Tropical Research Institute, Bocas del Toro, Panama June-Dec. 2013

Investigated the role of consumers in coral-algal community dynamics under the supervision of Dr. Andrew Altieri.

Peer-Reviewed Publications

I have 46 peer-reviewed publications (h-index 22), 17 as lead author, >1700 citations. International collaboration on ~85% of publications. See Web of Science, Google scholar or ORCID

49. Metcher, S., Stockbridge, J.S., **Kuempel,C.D.** (under review). Integrating sustainable bivalve aquaculture into water quality strategies to mitigate anthropogenic nitrogen loads. *Marine Pollution Bulletin*

48. Arnold, J., Chauvenet, A.M.,**Kuempel,C.D.** (under review). The conservation potential of Australia's marine protected area network. *Ecological Indicators*

47. Christofidis, M., Schoeman, D., Harte, J., Adams, V., Stockbridge, J., **Kuempel,C.D.** (under review). Climate analogues as a tool for marine aquaculture planning and adaptation *Earth's Future*

46. Davey, M., Klein, C.J., Roelfsema, C., **Kuempel,C.D.**, Possingham, H.P. (2025). > A review of the application of coral reef condition indicators in conservation planning. *Ecological Indicators*.

45. **Kuempel, C.D.**, Christofidis, M., Fuller, R., Bush, R., Garnett, S.T., Jackson, M.V. (2025). Half of coastal aquaculture sites in Queensland are in close proximity to shorebird habitats. *Ocean and Coastal Management*

44. Dominguez-Martinez, Klein, Blanchard, Farmery, **Kuempel, C.D.**, Roberson, Spillias, Cottrell (2025). Seafood Sustainability Challenges for Import-Dependent Nations *Reviews in Fisheries Science and Aquaculture*

43. Christofidis, M., Lyons, M., **Kuempel, C.D.** (2025). One-third of Australia's coastal terrestrial aquaculture at risk from sea level rise. *Aquaculture Science and Management*

42. Stockbridge, J.S., Brown, C.J., **Kuempel, C.D.** (2025). Mapping the global co-location potential of offshorewind energy and aquaculture production. *Ocean and Coastal Management* (IF = 4.3)

41. Vozzo, M., Christofidis, M. Griffiths, L., Kelly, R., Manion, M.L., Barmand, S., Bolin, J., Fuenzalida-Miralles, P., Harris, M., Issell, A., *Kuempel, C.D.*, Diaz, M., Murunga, M., Palmer, S., Schaefer, N., Simpson, J. (2025). Advancing ocean sustainability through better science integration: perspectives of Early Career Ocean Professionals *Front. in Ocean Sust.*

40. **Kuempel, C.D.**, Stockbridge, J.S., O'Neill, T., Griffiths, L., Frid,C.L.J., Brown, C.J. (2025). Frameworks and best practices for cumulative impact accounting in offshore wind energy development. *Marine Policy* (IF = 3.8)

39. Fleury, A., O'Hara, C.C., Butt, N., Restrepo, J., Halpern, B.S., Klein, C.J., **Kuempel, C.D.**, Gaynor, K.M., Bentley, L.K., Richardson, A.J., Dunn, D.C. (2024). Spatial and life history variation in a trait-based species vulnerability and impact model. *PlosOne* (IF = 3.8)

38. **Kuempel, C.D.**, Arnett, E., Klein, C.J. (2024). Quantifying global redundant fisheries trade to streamline seafood supply chains. *PlosOne* (IF = 3.8)

37. **Kuempel, C.D.** and Suarez-Castro, A.F. (2024) The Global Biodiversity Framework can be leveraged to better manage transboundary watersheds. *Proceedings of the National Academy of Sciences* (IF = 12)

36. **Kuempel, C.D.**, Thomas, J., Wenger, A., Jupiter, S.D., Suarez-Castro, A.F., Nasim, N., Klein, C.J., Hoegh-Guldberg, O. (2024). A spatial framework for improved sanitation to support coral reef conservation. *Environmental Pollution* (IF = 8.9)

35. Vella, T., Roberson, L., **Kuempel, C.D.**, Klein, C (2023). Quantifying the accessibility of sustainable seafood in South East Queensland, Australia. *Marine and Freshwater Research* (IF = 2.3)

34. Adams, V., Chauvenet, A.L.M., Stoudmann, N., Gurney, G.G., Brockington, D., **Kuempel, C.D.** (2023). Multiple-use protected areas are critical to equitable and effective conservation. *One Earth* (IF = 16.7)

33. Halpern, B.S., Frazier, M.,Rayner, P.E., Clawson, S.G., Blanchard, J.L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Jacobsen, N.S., **Kuempel, C.D.**, M., Moran, D., Nash, K.L., Williams, D.R. (2023). Matters Arising - Diving deeper into fisheries disturbance. *Nature Sustainability* (IF = 29.2)

32. Froehlich, H., Montgomery, J.C., Williams, D.R., O'Hara, C., **Kuempel, C.D.**, Halpern, B.S. (2023). Biological life history and farming scenarios of marine aquaculture help reduce wild marine fishing pressure. *Fish and Fisheries* (IF = 7.3)

31. Wakwella, A., Wenger, A., Jenkins, A., Lamb, J., **Kuempel, C.D.**, Claar, D., Corbin, C., Falinski, K., Rivera, A., Grantham, H.S., Jupiter, S.D. (2023). Integrated watershed management solutions for healthy coastal ecosystems and people. *Cambridge Prisms: Coastal Futures*

30. **Kuempel, C.D.**, Frazier, M., Verstaen, J., Rayner, P.E., Blanchard, J.L., Bouwman, L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Jacobsen, N.S., McIntyre, P.B., Metian, M., Moran, D., Nash, K.L., Tobben, J., Williams, D.R., Halpern, B.S. (2023). Feed drives land-sea overlap of the global cumulative environmental pressure of farmed chicken and salmon. *Current Biology* (IF = 11)

29. Beger, M., Metaxas, A., Balbar, A.C., McGowan, J.A., Daigle, R., **Kuempel, C.D.**, Treml, E.A., Possingham, H.P. (2022). Demystifying ecological connectivity for actionable spatial conservation planning. *Trends in Ecology and Evolution* (IF = 18.1 citations = 9)

28. Halpern, B.S., Frazier, M., Verstaen, J., Rayner, P.E., Clawson, S.G., Blanchard, J.L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Jacobsen, N.S., **Kuempel, C.D.**, McIntyre, P.B., Metian, M., Moran, D., Nash, K.L., Tobben, J., Williams, D.R. (2022). The cumulative environmental footprint of global food production. *Nature Sustainability* (IF = 29.2, citations

20)

27. Hicks, C.C. Gephart, J.A., Koehn, J.Z., Nakayama, S., Payne, H., Allison, E.H., Delhabib, D., Cao, L., Cohen, P.J., Fanzo, J., Fluet-Chouinard, E., Gelcich, S., Golden, C.D., Gorospe, K.D., Isaacs, M., **Kuempel, C.D.**, Lee, K.N., MacNeil, M.A., Maire, E., Njuki, J., Rao, N., Sumaila, U.R., Selig, E.R., Thilsted, S.H., Wabnitz, C.C.C., Naylor, R.L. (2022). Towards Justice in Blue Food Systems. *Nature Food* (IF = 23.2, citations 14)

26. Klein, C.J., **Kuempel, C.D.**, Watson, R.A., Teneva, L., Coll, M., Mora, C. (2022). Global fishing between jurisdictions with unequal fisheries management. *Environmental Research Letters* (IF = 7.8, citations 5)

25. Almaraz, M., **Kuempel, C.D.**, Salter, A., Houlton, B.Z., Halpern, B.S. (2022). The impact of excessive protein consumption on human wastewater nitrogen loading of US waters. *Frontiers in Ecology and the Environment* (IF = 12.2, citations 5)

24. Clawson, S.G.,**Kuempel, C.D.**, Blasco, G., Froehlich, H.E., Cottrell, R.S., Metian, M., Tobben, J., Williams, D.R., Nash, K.L., Frazier, M., Verstaen, J., Halpern, B.S. (2022). Mapping the spatial distribution of global mariculture production. *Aquaculture* (IF = 4.6, citations 12)

23. Williams, B.A., Watson, J.E.M., Beyer, H.L., Klein, C.J., Montgomery, J., Runting, R.K., Roberson, L.A., Halpern, B.S., Grantham, H.S., **Kuempel, C.D.**, Frazier, M., Venter, O., Wenger, A. (2021). The global rarity of intact coastal regions. *Conservation Biology* (IF = 6.5, citations 39)

22. **Kuempel, C.D.**, Simmons, B.A., Davey, M.(2022) Assessing he status of existing and tentative marine natural World Heritage areas reveals opportunities to better achieve World Heritage Convention goals. *Journal of Environmental Management* (IF = 8.4, citations 1)

21. **Kuempel, C.D.**, Tulloch, V.J.D., Giffin, A.L., Simmons, B.A., Hagger, V., Phua, C. and Hoegh-Guldberg, O. (2021). Identifying management opportunities to combat climate, land and marine threats across less climate exposed coral reefs. *Conservation Biology* (IF = 6.5, citations 2)

20. Roberson, L., Beyer, H.L., O'Hara, C., Watson, J.E.M., Dunn, Daniel, Klein, C.J., Frazier, M., **Kuempel, C.D.**, Williams, B., Grantham, H., Montgomery, J., Kark, S., Runting, R.K. (2021). Multinational coordination required for conservation of at least 90% of marine species. *Global Change Biology* (IF = 12.3, citations 16)

19. **Kuempel, C.D.**, Chauvenet, A.L.M., Symes, W.S., Possingham, H.P.P. (2021). Predicted protected area downsizing decreases habitat representation across terrestrial ecoregions in the tropics and subtropics. *Conservation Science and Practice* (IF = 3.3, citations 3)

18. Suarez-Castro, A.F., Beyer, H.L., **Kuempel, C.D.**, Linke, S., Borelli, P., Hoegh-Guldberg, O. (2021). Global forest restoration opportunities to foster coral reef conservation. *Global Change Biology* (IF = 12.3, citations 16)

17. Cottrell, R.S., Metian, M., Froehlich, H.E., Blanchard, J.L., Jacobsen, N.S., McIntyre, P.B., Nash, K.L., Williams, D.R., Bouwman, L., Gephart, J.A., **Kuempel, C.D.**, Moran, D.D., Troell, M., Halpern, B.S. (2021) Time to rethink trophic levels in aquaculture policy. *Reviews in Aquaculture* (IF = 11.1, citations
36) *Sena De Silva Award for novelty and impact*

16. **Kuempel, C.D.**, Froehlich, H.E., Halpern, B.S. (2021) An informed though experiment exploring the potential for a paradigm shift in aquatic food production. *Ocean and Coastal Management* (IF = 4.3, citations 6)

15. Adams, V.M., Dimitrova, N., Possingham, H.P., Allan, J.R., **Kuempel, C.D.**, Peterson, N., Kaiye, A. Keako, M., Tulloch, V.J.D. (2021) Scheduling incremental actions to build

a comprehensive national protected area netowrk for Papua New Guinea. *Conservation Science and Practice* (IF = 3.3, citations
7)

14. Chauvenet, A.L.M., Watson, J.E.M., Adams, V.M., Di Marco, M., Venter, O., Davis, K.J., Mapping, B., Klein, C.J., **Kuempel, C.D.**, Possingham, H.P. (2020). To achieve big wins for conservation, prioritise protection of ecoregions closest to meeting targets. *One Earth* (IF = 16.7, citations 17)

13. **Kuempel, C.D.**, Frazier, M., Nash, K.L., Sand Jacobsen, N., Williams, D.R., Blanchard, J.L., Cottrell, R.S., McIntyre, P.B., Moran, D., Bouwman, L., Froehlich, H.E., Gephart, J.A., Metian, M., Tobben, J., Halpern, B.S. (2020). Integrating life-cycle and impact assessments to map food's cumulative environmental footprint. *One Earth* (IF = 16.7, citations 13)

12. Daigle, R., Metaxas, A., Balbar, A., McGowan, J., Treml, E., **Kuempel, C.D.**, Possingham, H., Beger, M. (2020). Operationalizing ecological connectivity to identify spatial conservation priorities with Marxan Connect. *Methods in Ecology and Evolution*, (IF = 9.2, citations 87) *Top cited paper 2022*

11. **Kuempel, C.D.**, Chauvenet, A.L.M., Possingham, H.P., Adams, V.M. (2020). Evidence-based guidelines for prioritizing investments to meet international conservation objectives. *One Earth* (IF = 16.7, citations 16)

10. Halpern, B.S., Blanchard, J.L., Bouwman, L., Cottrell, R.S., Froehlich, H.E., Gephart, J.A., Sand Jacobson, N.,**Kuempel, C.D.**, McIntyre, P.B., Metian, M., Moran, D., Nash, K.L., Tobben, J., Williams, D.R. (2019). Opinion: Putting all food on the same table: Achieving sustainable food systems requires full accounting. *Proceedings of the National Academy of Sciences* (IF = 12, citations 74)

9. **Kuempel, C.D.**, Jones, K.J., Watson J.E.M., Possingham, H.P. (2019). Quantifying biases in placement of marine protected areas relative to abatable biodiversity threats. *Conservation Biology* (IF = 6.5, citations 39) *Top cited paper 2019*

8. Jantke, K., **Kuempel, C.D.**, McGowan, J.M., Chauvenet, A.L.M., Possingham, H.P. (2018). Metrics for evaluating representation target achievement in protected area networks. *Diversity and Distributions* (IF = 5, citations 24)

7. Jones, K.J., Klein, C.J., Halpern, B.S., Venter, O., Grantham, H., **Kuempel, C.D.**, Shumway, N., Friedlander, A.M., Possingham, H.P., Watson, J.E.M. (2018). The location and protection status of Earth's diminishing marine wilderness. *Current Biology* (IF = 11, citations 240)

6. **Kuempel, C.D.**, Adams, V.M., Possingham H.P., Bode M. (2018) Bigger or better: the relative benefits of protected area network expansion and enforcement for the conservation of an exploited species. *Conservation Letters* (IF = 8.3, citations 41)

5. Campbell, J.E., Altieri, A.H., Johnston, L., **Kuempel, C.D.**, Paperno, R., Paul, V.J., Duffy, J.E. (2018) Herbivore community determines the magnitude and mechanism of nutrient effects on subtropical and tropical seagrass. *Journal of Ecology* (IF = 6.3, citations 36)

4. Chauvenet, A.L.M., **Kuempel, C.D.**, McGowan J., Beger, M., Possingham, H.P. (2017). Methods for calculating Protection Equality for conservation planning. *PloS one* (IF = 3.8, citations 40)

3. **Kuempel, C.D.** and Altieri A. H. (2017). The emergent role of small-bodied herbivores in pre-empting phase shifts on degraded coral reefs. *Scientific Reports* (IF = 4.9, citations 58)

2. **Kuempel, C.D.**, Chauvenet, A.L.M., and Possingham H.P. (2016). Equitable representation of ecoregions is slowly improving despite strategic planning shortfalls. *Conservation Letters* (IF = 8.3, citations 41)

1. Nelson, H.R., **Kuempel, C.D.**, and Altieri, A.H. (2016). The resilience of reef invertebrate biodiversity to coral mortality. *Ecosphere* (IF = 3.4, citations 50)

Other publications, reports and white papers

7. Griffiths, Leaning, Cossu, Dunstan, Haward, **Kuempel, C.D.**, Lacharite, Marsh, Stockbridge, Woolley, Vince, Frid (2025). Key enablers for implementation of the Australian Marine Spatial Planning Framework. *Blue Economy Cooperative Research > Centre*
6. Griffiths, Breakey, Bossi, Cossu, Dunstan, Hasan,**Kuempel, C.D.**, Lacharite, Marsh, Skinsley, Smart, Smith, Stockbridge, Woolley, Frid (2025). Implementing an Australian Marine Spatial Planning Framework. *Blue Economy Cooperative Research Centre*
5. Griffiths, Cossu, Dunstan, Haward, **Kuempel, C.D.**, Lacharite, Leaning, Marsh, Stockbridge, Woolley, Frid (2025). Guiding principles for an Austrlaian MSP Framework. *Blue Economy Cooperative Research Centre*
4. Stockbridge and **Kuempel, C.D.** (2024). Identifying synergies and trade-offs between sectors in the Blue Economy Zone *Blue Economy Cooperative Research Centre*
3. Stockbridge and **Kuempel, C.D.** (2024). Evaluation of potential for interaction and/or conflict among ocean users and the environment in offshore waters in Australia *Blue Economy Cooperative Research Centre*
2. **Kuempel, C.D.** (2023) Sedimentation sifted out of global pollution priorities. *Science* (IF = 54.5)
1. Wenger, Jaurez, Thomas, Amaya, Corbin, Edmond, Falinski, Hill, Jenkins, Jupiter, **Kuempel, C.D.**, Lamb, Nalley, Omwenga, Oza (2023). A guide for integrated conservation and sanitation programs and approaches.*Wildlife Conservation Society*

Grants* and consulting†

DFAT Australia Awards Nature-based solutions short course in Papua New Guinea ($167k) †	(2025)
WWF modelling sediment run-off for Fiji's Great Sea Reef ($40k) †	(2025)
Bush Heritage Bespoke R Training Course ($16.5k) †	(2024)
Inspiring Queensland National Science Week Grant (Brisbane Soapbox Science) † ($4k)	(2024)
WWF Green Climate Fund Fiji Feasibility Study ($55k) †	(2023-2024)
SPREP Kiwa Initiative Micro-credentials ($135,335) †	(2024)
Future Earth Australia Creating Opportunities Grant ($6k)*	(2023)
Griffith Advancing Women's Research Success Grant ($4,601.73)*	(2023)
Blue Economy CRC Marine Spatial Planning ($1.3M, responsible for $270k in post-doc salary)*	(2023)
Global Environmental Facility Coral Reef Rescue project (US$7M)*	(2022)
CBCS Small Grants Scheme ($4,500 across three successful proposals)*	(2021)
ARC Laureate Postgraduate Scholarship ($119,376 over 4 years)	Nov 2014-2018
Seafood Footprint Displacement Top-up Scholarship (Dr. Carissa Klein, $5,000)	March-June 2018
50 Reefs Top-up Scholarship ($5,000)	Sept. 2017
University of Queensland Career Development Award ($5,000)	May 2017
Great Barrier Reef Marine Park Authority Science for Management Award ($1,000-declined)	May 2017
Australian Coral Reef Society Writing Retreat Scholarship ($500)	May 2016

Smithsonian Tropical Research Institute Short-Term Fellowship ($2,000)* June-Nov. 2013

University of Iowa Old Gold Scholarship ($12,000) Aug. 2007-May 2011

University of Iowa Athletic Scholarship (Rowing) Jan. 2008-May 2011

Select teaching experience

3606ENV Conservation Biology - course convener, lecturer

> ~110 students (T1 2023 and 2024), SEC/SET >4 with feedback that I am "knowledgeable and excited about the content of the course and presents lecture material in a way that makes it easy to learn."

3804ENV Marine Ecosystem Modelling - course convener, lecturer

> ~30 students (T1 2023 and 2024), SEC/SET >4 with feedback that "I loved this course! The material I found very interesting and challenging. This course made me realise this is the avenue I want to go down after graduation"

3033ESC Environmental Data Analytics - lecturer

> 4 students (T2 2023), SEC/SET 5 (1 response) with feedback that "Caitie's teaching was great, she was patient and always willing to help. She would often push us to figure out things by giving hints to help guide you without just giving away the answer which I really liked. Definitely one of the best studio/tutorial teachers I've had during my degree."

Supervision (* denotes primary supervisor)

Ainsley Leaning, PhD (2025 - present)

> Ecosystem characteristication for marine spatial planning using hydroacoustics

Jade Arnold*, Masters (2024 - first class)

> The conservation potential of Australia's marine protected area network

Simon Metcher*, current PhD, former Honours Student (2024-current) > PhD focused on the impacts of aquaculture production in Australia (domestic > scholarship recipients) > B.S. (Hons) Assessing bivalve farming potential to mitigate nutrient pollution in the > Great Barrier Reef

Siaan Charles*, Advanced Studies (2024)

> Methods for First nations engagement in Marine Spatial Planning

Dr Jackson Stockbridge*, Post-doctoral fellow (2023-present)

> Developing a framework for assessing cumulative impacts of the blue economy in Australia

Marina Christofidis*, PhD Student (2023-present)

> Climate risk to aquaculture production in Australia (international scholarship recipient)

Madeline Davey , PhD (2022 - present)

> PhD focused on metrics and tools for better conservation planning on coral reefs

Alana McPherson, Honours Student (2023-2024)

> Conservation benefits of Marine Protected Areas over time in Australia

Sophie Petrie*, Capstone project, (2023)

> The Potential Impacts of Aquaculture on Threatened Species in Queensland

Emma Arnett, Master's Student (2020-2021)

> Quantifying like-for-like global seafood trade Now employed at DETSI

Ella Sinclair, Volunteer (2020)

Primary supervisor for conducting literature review of monitoring and evaluation indicators for WWF Coral Reef Rescue Initiative Now employed as an environmental consultant

Awards

Queensland Women in STEM Award Finalist (1 of 14 finalists) (2025)

Vice Chancellor's Research Excellence Award – Early Career Researcher nominee (1 of 4 nominees) (2023)

Foundation for the WA Museum Fame Lab Semi-finalist: To lower your environmental footprint consider what your dinner ate (2023)

University of Queensland Faculty of Science Awards Rising Star - Academic category Finalist (Dec. 2020)

APEC Science Prize for Innovation, Research and Education Award - shortlisted for Australian nominee (2020)

University of Queensland Dean's Award for Outstanding HDR Theses (2018)

Taylor and Francis University of Queensland Library Excellence Award in Research ($1,000) May 2017

Select Media and Outreach

As an example of reaching a broad non-scientific audience, my written work for The Conversation blog (theconversation.com) has had >200,000 readers.

Queensland prawn areas at risk of inundation by rising seas this century The Conversation

Which food is better for the planet? The Washington Post

Food for thought: the carbon footprint of salmon and chicken farming mostly stems from feed. The Guardian

To lower food emissions consider what your dinner ate. Popular Science

Farmed chicken and salmon have a global footprint. Nature

Coming out or not, earthquake aid, and what's the most eco-friendly meat? Triple J Hack

Female scientists encouraging children to embrace science Courier Mail

Select presentations

* *oral presentation*, † invited presentation

Kuempel C† (2024) Frameworks to assess cumulative impacts of Australia's Blue Economy *International Marine Conservation Congress, Cape Town, South Africa*

Kuempel C† (2024) The need and feasibility of the Fiji Coral Reef Resilience project *International Marine Conservation Congress, Cape Town, South Africa*

Kuempel C† (2024) Frameworks to assess cumulative impacts of Australia's Blue Economy *Australian Marine Science Association*

Kuempel C† (2023) Frameworks to assess cumulative impacts of Australia's Blue Economy *Blue Economy CRC Participants Workshop, Kingscliff, Australia*

Kuempel C† (2023) Putting Australian Aquaculture on the Map *Australian Marine Sciences Association, Gold Coast, Australia*

Kuempel C† (2022) Mapping a path for conservation in a developing world. *Griffith University, Brisbane, Australia*

Kuempel C† (2022) Conservaiton nature in a developing world. *Linn-Mar High School, Marion, IA, USA*

Kuempel C (2021) Informing management across less-climate exposed coral reefs. *International Congress on Conservation Biology*, on-line

Kuempel C† (2021) Coral Reef Rescue Initiative: A climate-centric approach for managing coral reefs for nature and people. *Coral Reef Rehabilitation and Management Program - Coral Triangle Initiative*, on-line

Current service and Outreach Roles

Academic

New Zealand Endeavour Fund Offshore wind working group - international advisory group

International Congress on Conservation Biology 2025 *Steering Committee Member*

School of Environment and Science *Early Career Representative* (2024-present)

Australian Rivers Institute *Early Career Researcher Representative* (2023)

Griffith Women in Leadership 2023 Cohort

Centre for Biodiversity and Conservation Science Management Committee *Early Career Researcher Education Lead* (2021-Present)

UNEP MPA Toolbox and Learning Platform *Academic Associate, Content editor/coordinator/contributor*

Reviewer for international publications

> Nature Sustainability, the Journal of Environmental Management, Biological Conservation, Marine Biology, PLoS One, Global Ecology and Conservation, Coral Reefs, Conservation Letters, Conservation Biology, Diversity and Distributions, Marine Policy, Conservation Science and Practice and PeerJ.

Community

Soapbox Science Brisbane co-organiser (2024-present) - community engagement to promote women in STEM during National Science Week

Certified Software Carpentries Instructor (2020-present) - specialising in introductory to advanced R programming skills and version control (e.g., GitHub).

RLadies Brisbane Oraganiser - promoting women in programming (2020-2024)

Marxan Spatial Planning Instructor (2017-current) - successful workshops in Australia, Indonesia, Colombia, and online

Professional Membership

Australian Marine Sciences Association

Reef Life Survey Volunteer (Coral Sea expedition 2018)

Australian Coral Reef Society

Society for Conservation Biology > *Board Member/Treasurer* (2015-2017)

International Society for Reef Studies

World Commission on Protected Areas Young Professionals Group

Working Groups

Organiser/Participant Australia's sustainable seafood future, UQ Heron Island Research Station, April 2022

Organiser/Participant Centre for Biodiversity and Conservation Science HDR/ECR Writing workshop, Moreton Bay Research Station, Dec. 2021

Participant Science for Nature and People Partnership (SNAPP) Coastal Health working group, NCEAS (on-line), Nov. 2021

Organiser/Participant CBCS Integrated Watershed Management Workshop, University of Queensland, Oct. 2021

Panel Participant Working with "big data" in the biological sciences, University of Queensland, July 2021

Organiser/Participant Coral Reef Rescue Initiative Near Real-Time Monitoring Workshop, WWF online, Sept. 2021

Participant Strengthening coastal communities resilience by supporting community enterprises, WWF online, Oct. 2020

Participant What's the catch? Social and environmental sustainability of seafood, Heron Island, University of Queensland Jan 2021

Post-doctoral Fellow Comparing the environmental impacts of global food systems, NCEAS, UC Santa Barbara Sept. 2018-2020

Participant Sustainable seafood and marine conservation, Heron Island, University of Queensland June 2019

Participant Incorporating connectivity into spatial conservation planning,The Nature Conservancy, Washington D.C. 2017

Participant/Coordinator Land-sea conservation planning in Papua New Guinea, Port Moresby, Papua New Guinea 2017

Participant/Coordinator Marine spatial planning in the Sunda Banda Seascape, Bali, Indonesia 2016

Organizer/Participant Global Marine Conservation. Lady Elliot Island, Australia 2016

Skills

Programming: = R (*advanced*), Github (*advanced*), Matlab (*basic*) *Software:* = Marxan (*advanced*), Arc/QGIS, Adobe Illustrator, Microsoft Office